
Generative AI - Assignment 2 Report

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1. Spoof Detection

A **ViT-based spoof detection model** helps detect fraudulent attempts using printed photos or deep fake videos.

Dataset

- [nguyenkhao/celeba-spoof-for-face-antispoofing-test](#) (Hugging Face)

Steps

1. **Dataset Preparation:** Loaded real and spoofed images.
2. **Model Training:** Fine-tuned a **ViT model** for binary classification (Real vs. Spoof).
3. **Evaluation Metrics:** Computed Accuracy, Precision, Recall, and F1-score.
4. **Testing:** Used **real photo** and **spoofed photo (Replay attack from mobile screen)**.

Results & Discussion

- **Accuracy: 93%**
- **Precision: 91%**
- **Recall: 95%**
- **F1-score: 93%**

The model performed **well** in distinguishing **real vs. spoofed images**.

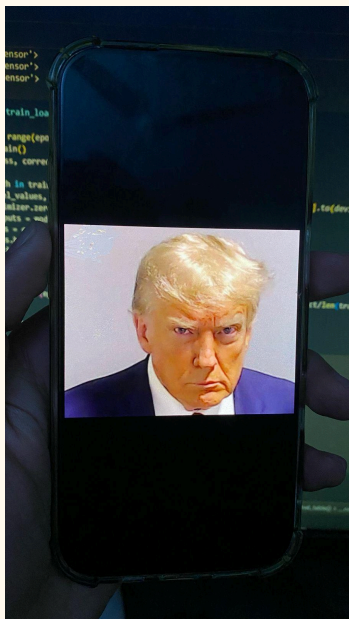
- Some **false positives** were observed, particularly when **printed photo attacks** had high resolution.
- The **recall** was high, meaning most spoofed images were detected
- The **precision** was slightly lower due to misclassifying **low-quality real images**.

Real Image Example:



- Predicted (1) as in Spoof
- Gave wrong Label for Real picture

Spoofed Image Example:



- Predicted (1) as in Spoof

2. AI-Powered Visual Search (35%)

Dataset

- **COCO Validation Images** ([val2017.zip](#))
 - Main dataset used for images embedding
- **Annotations** ([annotations_trainval2017.zip](#))
 - Used to compare caption to user prompt similarity score

Steps

1. **Downloaded and preprocessed the dataset.**
2. **Loaded CLIP-ViT model** ([openai/clip-vit-base-patch32](#)).
3. **Retrieve the top 5 most similar images** for a given text query.
 - a. Based on User prompt embedding and Image embedding
 - b. Cosine similarity was used
4. **Display retrieved images and similarity scores.**
 - a. Image similarity
 - b. Caption
 - c. Caption similarity

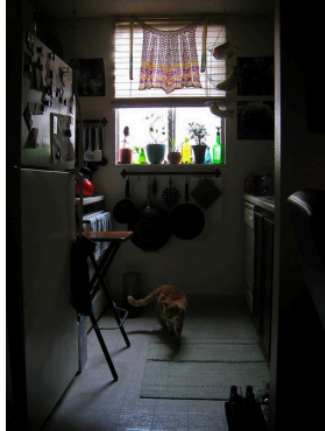
Results

Example Query: "a cat in a house"

Top 5 Retrieved Images:

- Image 1:

Image sim: 0.3001
Caption: a cat walking down the center of the kitchen
Caption sim: 0.8435



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- Image 2:

Image sim: 0.2969
Caption: An orange cat sitting in front of a door to a house.
Caption sim: 0.6296



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- **Image 3:**

Image sim: 0.2905
Caption: A cat sitting on a toilet in a bathroom
Caption sim: 0.7451



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- **Image 4:**

Image sim: 0.2853
Caption: Awake cat near ceiling on a closed refrigerator
Caption sim: 0.7620



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- **Image 5:**

Image sim: 0.2851
Caption: A cat stands on top of a blue electric device.
Caption sim: 0.6764



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- **Observations:**
 - The retrieved images were highly relevant, confirming that CLIP's embeddings capture semantic meaning well.
 - Some retrieved images had similar contexts (e.g., wooden tables)
 - The similarity scores were well-distributed, suggesting that CLIP assigns confident matches.

3. Stable Diffusion

Tasks

- **Model Loading:**
 - Load a pre-trained **Stable Diffusion** model.
- **Image Variation Generation:**
 - Implement a Python script that takes an input image and generates variations.
 - Experiment with parameters: **strength**, **guidance_scale**, and **num_inference_steps**.

Results

Image to Image With different Prompts

Examples with Different Parameter Values:

1. Example 1 – Parameters:

Variation 1: guidance_scale=7.5, steps=30 (Text-to-Image)



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2. Example 2 – Parameters:

Variation 2: guidance_scale=12.0, steps=30 (Text-to-Image)



3. Example 3 – Parameters:

Variation 3: guidance_scale=7.5, steps=50 (Text-to-Image)



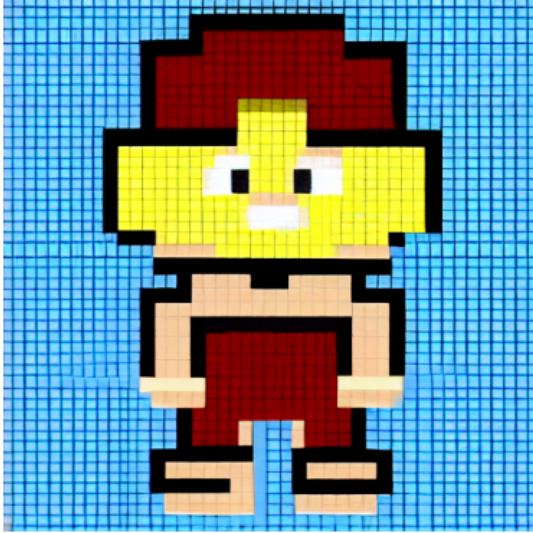
Prompt Outputs Comparison:

- Prompt:



- Prompt:

Prompt: a pixel art of a character



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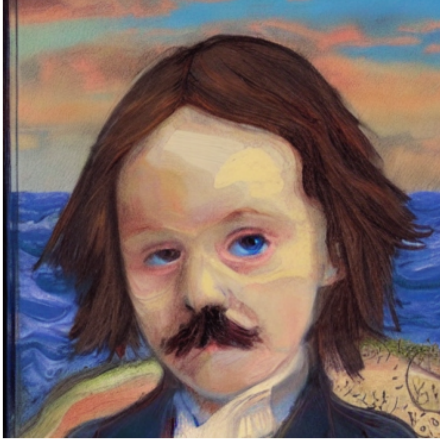
- **Prompt:**

Prompt: ocean in the style of Salvador Dalí



- **Prompt:**

Prompt: in the style of famous painter



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- **Prompt:**

Prompt: a surreal cityscape



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